

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

50% Time Horizon **doubles every ~7 months**

From GPT-2 (2 sec) → o3 (~2 hours) in 6 years

A New Metric: 50% Task-Completion Time Horizon

The task duration (in skilled-human time) at which an AI model achieves a 50% success rate — providing an intuitive, human-grounded measure of autonomous capability.

1 Assemble Tasks

170 diverse software
& research tasks



2 Measure

Human timing +
AI success rates



3 Fit Model

Logistic regression
→ 50% threshold

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

HCAST

97 tasks

Diverse software tasks | 1 min – 30 hrs

RE-Bench

7 tasks

ML research engineering | 8 hrs each

SWAA

66 tasks

Single-step atomic actions | 1 sec – 30 sec

Task Family	Human Time	Description
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oxdna_simple	9 minutes	Detect and fix bug in molecular dynamics simulation input files
munge_data	56 minutes	Write Python script to transform JSON data by inferring conversion rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of backtesting tool

800+ human baselines | 2,529 hours of human timing data collected

Exponential Trend: 2 Seconds → ~2 Hours in 6 Years

~207 days

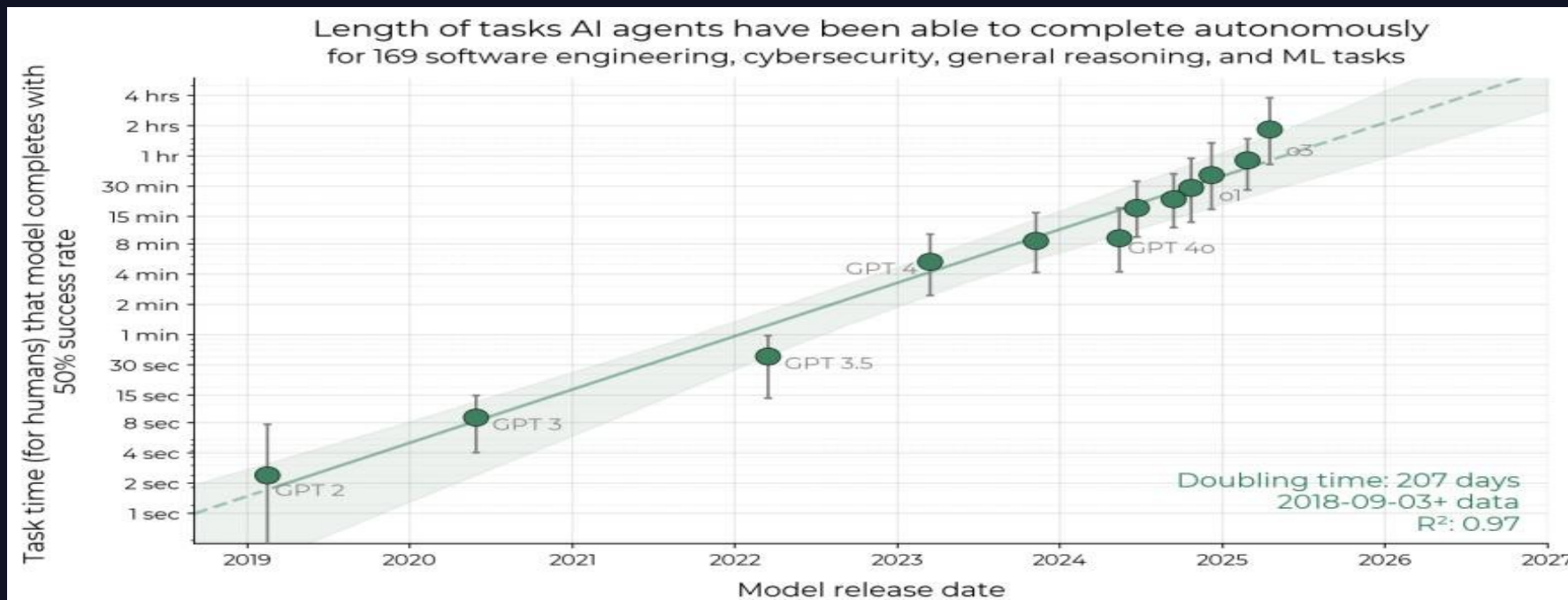
Doubling Time

$R^2 = 0.97$

Fit Quality

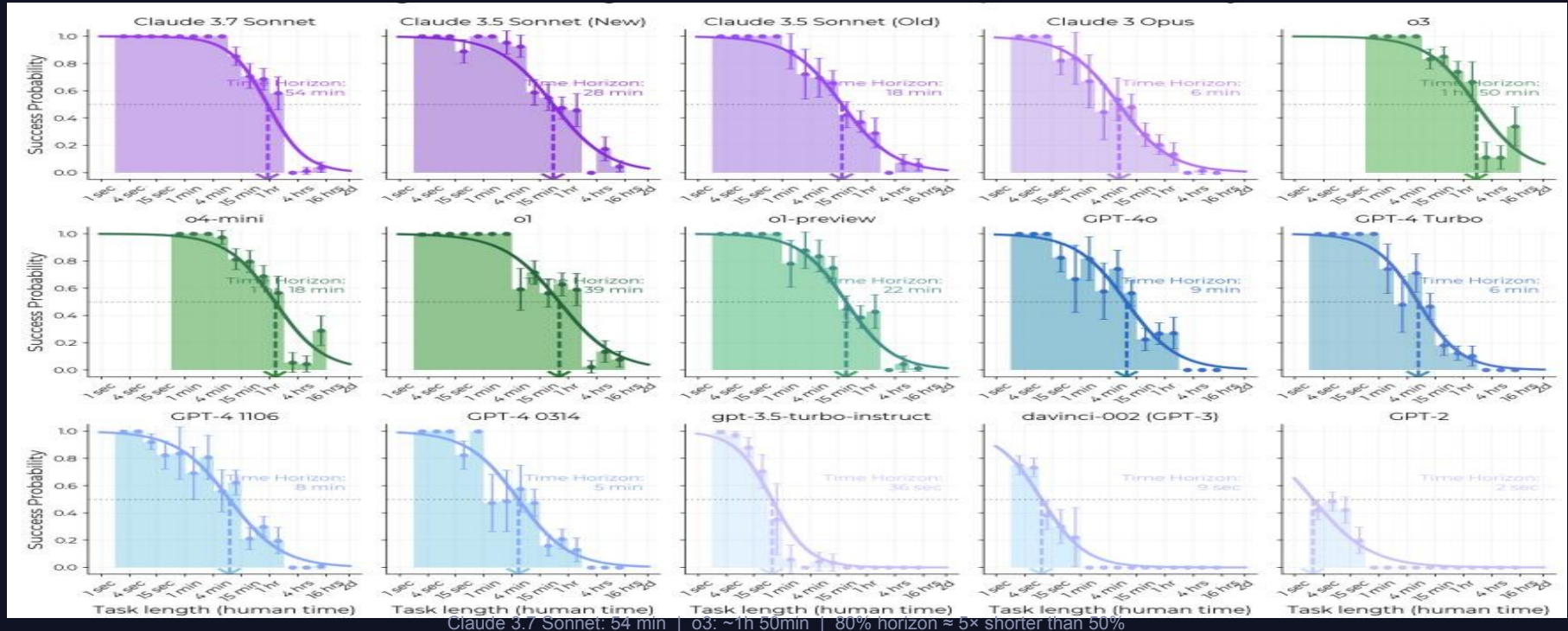
6 years

GPT-2 → o3



Success Probability Drops Sharply with Task Length

Each curve shows one model's success rate vs. human-time task duration. Frontier models push the curve rightward — succeeding at longer tasks.



Reasoning, Tool Use, and Error Recovery Drive Growth



Improved Logical Reasoning

Models demonstrate stronger multi-step reasoning chains and planning ability across complex software tasks.



Better Tool Use

More effective use of code execution, file manipulation, and external tools in agentic workflows.



Reliability & Self-Awareness

Greater consistency in task execution and ability to adapt when encountering mistakes or dead ends.



Limitation: Messy Tasks

Performance remains notably lower on less structured, ambiguous tasks that lack clear specifications.

Extrapolation: Month-Long Task Automation by 2028–2031

167

work hours
(~1 month)



2025
~2 hrs



2028
Earliest



2031
Latest

2023–2025 growth rate is ~20% faster than the overall 2019–2025 rate, suggesting possible acceleration.

Naive extrapolation of the exponential trend places the 1-month time horizon (167 work hours) between mid-2028 and mid-2031.

External validity remains an open question — supplementary experiments found little evidence that trends are slower on more realistic tasks.

Takeaways: Exponential Growth Demands Proactive Safety Governance

- 1** 50% time horizon is a robust, intuitive metric for tracking AI autonomy — $R^2 = 0.97$ over 6 years of model releases.
- 2** Doubling every ~7 months with possible acceleration since 2024 — the pace may be increasing.
- 3** Current frontier at ~2 hours of human-equivalent autonomous work (o3, early 2025).
- 4** External validity remains open — tasks may not perfectly represent real-world intellectual labor distributions.
- 5** Key limitations: poor performance on unstructured tasks; uncertain generalization to real-world job distributions.